

```
// OSSP Skill 21 - Module Integration and Runtime Error Handling
// Demonstrates modular-style functions, interface validation,
// syntax checking, runtime errors, messages, logging, and recovery.
```

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <errno.h>
#include <unistd.h>
#include <sys/wait.h>
```

```
static void log_error(const char *message) {
    FILE *log = fopen("skill21_errors.log", "a");

    if (log) {
        fprintf(log, "%s\n", message);
        fclose(log);
    }
}
```

```
static int validate_command(const char *cmd) {
    if (!cmd || *cmd == '\0') {
        fprintf(stderr, "Syntax error: empty command.\n");
        log_error("Empty command");
        return 0;
    }
}
```

```
if (strstr(cmd, "|")) {
    fprintf(stderr, "Syntax error: invalid pipeline operator.\n");
}
```

```
    if (strstr(cmd, "||")) {
        fprintf(stderr, "Syntax error: invalid pipeline operator.\n");
        log_error("Invalid pipeline syntax");
        return 0;
    }

    return 1;
}

static int run_command(const char *cmd) {
    pid_t pid = fork();

    if (pid == -1) {
        perror("fork");
        log_error("fork failed");
        return -1;
    }

    if (pid == 0) {
        execl("/bin/sh", "sh", "-c", cmd, (char *)NULL);
        perror("exec");
        _exit(127);
    }

    int status;
    if (waitpid(pid, &status, 0) == -1) {
        perror("waitpid");
        log_error("waitpid failed");
        return -1;
    }
}
```

```

    return -1;
}

if (WIFEXITED(status) && WEXITSTATUS(status) != 0) {
    char msg[256];
    snprintf(msg, sizeof(msg), "Command failed with status %d: %s",
             WEXITSTATUS(status), cmd);
    fprintf(stderr, "%s\n", msg);
    log_error(msg);
}

return 0;
}

```

```

int main(void) {
    char line[512];

    printf("=== Skill 21: Integration + Error Handling ===\n");
    printf("Try: echo Hello | wc -c\n");
    printf("Try an invalid command to see recovery.\n");
    printf("Type 'exit' to quit.\n\n");

    while (1) {
        printf("integrated> ");

        if (!fgets(line, sizeof(line), stdin))
            break;

        line[strcspn(line, "\n")] = '\0';
    }

```

```

        if (strcmp(line, "exit") == 0)
            break;

        if (!validate_command(line))
            continue;

        run_command(line);
    }

    printf("Program ended gracefully. Check skill21_errors.log for failures.\n");
    return 0;
}

```

```

sai_pavani@Sai-Pavani:~/OS_SKILL$ nano skill_21.c.c
sai_pavani@Sai-Pavani:~/OS_SKILL$ gcc skill_21.c.c -o skill_21
sai_pavani@Sai-Pavani:~/OS_SKILL$ ./skill_21
=== Skill 21: Integration + Error Handling ===
Try: echo Hello | wc -c
Try an invalid command to see recovery.
Type 'exit' to quit.

integrated> echo Hello | wc -c
6
integrated> ls
shellforge                skill4.c                skill_14.c.c            skill_20.c.c
shellforge.c              skill5                  skill_15                skill_21
simplecommandloop          skill5.c                skill_15.c.c            skill_21.c.c
simplecommandloop.c        skill_10                skill_16                skill_22.c.c
skill13_input.txt          skill_10.c.c            skill_16.c.c            skill_23.c.c
skill13_output.txt         skill_11                skill_17                skill_24.c.c
skill14_append.txt         skill_11.c.c            skill_17.c.c            skill_8
skill14_error.txt          skill_12                skill_18                skill_8.c.c
skill15_combined.txt       skill_12.c.c            skill_18.c.c            skill_9
skill3                     skill_13                skill_19                skill_9.c.c
skill3.c                   skill_13.c.c            skill_19.c.c            temp
skill4                     skill_14                skill_20

integrated> ps
  PID TTY          TIME CMD
   325 pts/0    00:00:00 bash
  1497 pts/0    00:00:00 skill_19
  1498 pts/0    00:00:00 skill_19
  1696 pts/0    00:00:00 skill_21
  1709 pts/0    00:00:00 sh
  1710 pts/0    00:00:00 ps

integrated> tr
tr: missing operand
Try 'tr --help' for more information.
Command failed with status 1: tr
integrated> help
sh: 1: help: Permission denied
Command failed with status 127: help
integrated> exit
Program ended gracefully. Check skill21_errors.log for failures.
sai_pavani@Sai-Pavani:~/OS_SKILL$ |

```